

CIE

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TITLE: **TradeLearn**

SCHOOL: Chitkara University, Rajpura, Punjab

**DETAIL OF THE STUDENT/MENTOR WHO UPLOADED THE COPYRIGHT ON GOVT PORTAL NEED TO
SHARE LOGIN AND PASSWORD WITH MENTORS**

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ABSTRACT

1.1 Problem Background:

In the modern digital era, skill development has become one of the most important factors for academic success, employability, and long-term career growth. The rapid advancement of technology and changing industry requirements continuously create demand for new skills. Students and professionals are expected not only to gain theoretical knowledge but also to acquire practical skills such as programming, communication, design, data analysis, leadership, and many others. Learning platforms available on the internet have significantly transformed education by providing access to courses and learning materials from anywhere in the world.

However, despite the availability of thousands of online learning resources, accessibility remains a major challenge. Most popular learning platforms rely on subscription-based models or require users to pay course fees. While these platforms provide quality educational content, they often become financially inaccessible for students, beginners, and individuals from economically limited backgrounds. The cost of acquiring multiple skills through paid platforms can become a burden, creating barriers to continuous learning.

At the same time, many individuals already possess valuable knowledge and practical skills that can benefit others. For example, one student may be skilled in web development while another may excel in graphic design, communication skills, mathematics, or digital marketing. Despite possessing useful expertise, there are limited systems that allow these individuals to exchange knowledge directly in a structured and mutually beneficial way.

Existing educational platforms primarily follow a one-directional learning approach where users consume content passively. Learners act as recipients rather than contributors. Knowledge sharing and teaching are often recognized only when monetary transactions are involved. This limits opportunities for collaborative learning and community-driven education.

Additionally, many individuals are willing to teach others but do not receive any meaningful incentive or recognition for their contribution. As a result, valuable knowledge remains underutilized and isolated within individuals instead of benefiting a broader community.

Therefore, there exists a need for a platform that promotes active participation, values knowledge as an exchangeable asset, and removes financial barriers associated with learning. The proposed system, TradeLearn, aims to address this challenge through a peer-to-peer skill exchange platform where learning and teaching become mutually rewarding activities. This creates a clear need for a platform that encourages mutual learning, values knowledge as an asset, and enables people to both teach and learn without financial barriers. TradeLearn is designed to address this gap by supporting peer-to-peer skill exchange through a transparent and structured system.

1.2 Objective of the Project

The primary objective of TradeLearn is to design and develop a web-based platform that facilitates peer-to-peer skill exchange through a structured credit-based learning system. The platform aims to create an environment where users can both teach and learn by exchanging skills rather than relying on monetary transactions. The system is intended to create an accessible and collaborative learning environment where knowledge becomes a valuable resource that can be exchanged among users.

One of the major objectives of TradeLearn is to promote collaborative learning by encouraging users to actively participate as both learners and educators. Unlike traditional platforms where users passively consume educational content, this platform enables individuals to contribute their own knowledge and engage in interactive learning experiences. This creates a mutually beneficial ecosystem where users can grow together by sharing and acquiring skills.

Another important objective is to remove financial barriers associated with skill development. Many students and early professionals are unable to afford expensive online courses and subscription-based learning platforms. TradeLearn addresses this issue by introducing a credit-based exchange mechanism that replaces monetary payments with earned learning credits, making education more accessible regardless of a user's financial background.

The platform also aims to encourage knowledge sharing by recognizing teaching as a valuable contribution. Users who teach others are rewarded with credits that can later be used

to learn new skills from other members of the community. This approach motivates individuals to actively participate and share their expertise rather than limiting themselves to being learners only.

TradeLearn further seeks to provide a structured exchange mechanism where users can create skill exchange requests, accept or reject requests, monitor learning progress, and maintain records of completed activities. This systematic process improves organization and transparency while ensuring smooth interaction between users.

Additionally, the project aims to build a community-driven educational ecosystem where users support one another throughout their learning journey. By encouraging collaboration and active participation, the platform helps create stronger learning communities and promotes peer-to-peer education.

Finally, another important objective of this project is to implement practical concepts of full-stack web development. Through the development of TradeLearn, various technical concepts such as frontend development, backend architecture, database management, authentication systems, API integration, and deployment strategies are implemented in a real-world application environment. This contributes not only to solving a practical problem but also to enhancing technical understanding and development skills.

1.3 Proposed Solution / System Overview

TradeLearn is proposed as a web-based peer-to-peer learning platform designed to enable skill exchange among users through a structured credit-based mechanism. Unlike traditional educational platforms that rely primarily on financial transactions, TradeLearn considers knowledge and teaching contributions as valuable assets. The platform aims to create an ecosystem where users can actively participate as both learners and educators, promoting collaborative and mutually beneficial learning experiences.

The workflow of the system begins when users register on the platform and create personalized profiles. During profile setup, users provide details about the skills they already possess and the skills they wish to learn. This information helps create a personalized learning environment and enables users to connect with individuals who can fulfill their learning requirements.

Once the profile creation process is completed, users can search and browse through available skills offered by other members on the platform. If a user finds another member who possesses expertise in a desired skill, they can initiate the interaction by sending a learning request. The receiving user can review the request details and make a decision based on availability and preferences. They may accept the request, reject it, or review additional information before responding. If the request is accepted, the learning process can begin and both users can proceed with the skill exchange activity.

TradeLearn introduces a credit-based mechanism that acts as an alternative to traditional payment systems. Users earn credits by teaching others and contributing their knowledge to the community. These earned credits can later be used to learn new skills from other users on the platform. In this system, credits function as a virtual currency that rewards active participation and encourages continuous engagement.

The platform also provides a dashboard that allows users to monitor and manage their activities efficiently. The dashboard displays important information such as learning history, skill exchange requests, available credits, pending activities, and exchange progress. Additionally, the system manages user authentication, authorization, request status tracking, and overall interactions to ensure secure, organized, and smooth platform functionality.

1.4 Key Features or Functionality

- **User Registration and Login:** Secure authentication for users to access the platform.
- **Profile Management:** Users can add skills they offer and skills they want to learn.
- **Skill Exchange Requests:** Users can send, accept, or reject learning requests.
- **Credit-Based System:** Users earn credits by teaching and use credits to learn.
- **Dashboard:** View requests, learning history, and available credits.
- **Request Tracking:** Monitor the status of each exchange (pending, accepted, completed).
- **Search and Browse Skills:** Find users based on the skills they offer.

1.5 Technology / Method Used

TradeLearn follows a full-stack web development architecture that integrates frontend technologies, backend technologies, database systems, authentication mechanisms, and development tools to build a complete and scalable web application. The selected technology stack ensures efficient communication between various system components while providing reliability, flexibility, and maintainability. The architecture is designed to support seamless interaction between users and the system while ensuring secure and efficient management of data.

The frontend of the application is developed using HTML, CSS, and EJS technologies. HTML is used to create the structural framework of web pages and define elements such as forms, buttons, navigation sections, and content layouts. CSS is responsible for handling the visual presentation of the platform by managing styling, layout design, responsiveness, colors, and user interface enhancements. It helps create an attractive and user-friendly experience across different devices and screen sizes. Embedded JavaScript (EJS) is used as the template engine for dynamic server-side rendering. EJS allows reusable templates and enables dynamic content generation based on user data. Its implementation reduces code redundancy, improves maintainability, and simplifies page rendering.

The backend of TradeLearn is developed using Node.js and Express.js. Node.js provides a server-side JavaScript runtime environment that enables efficient execution of application logic. One of the major advantages of Node.js is its non-blocking and event-driven architecture, which supports handling multiple user requests simultaneously. This improves system performance and scalability. Express.js is used as a backend framework to simplify server-side development. It provides features such as route handling, middleware integration,

request processing, and API development, making backend implementation more organized and manageable.

MongoDB is used as the primary database for storing application data. MongoDB is a document-oriented NoSQL database that offers flexibility and scalability for modern web applications. It was selected because of its flexible schema design, easy integration with JavaScript-based technologies, and ability to handle dynamic data structures efficiently. The database stores important information such as user profiles, offered skills, learning requests, credit records, authentication details, and exchange history.

For authentication and security purposes, JWT (JSON Web Token) is implemented in the system. JWT enables secure user login and verification by generating tokens that authenticate users during system interaction. It also helps manage sessions efficiently while maintaining security across protected routes and resources.

Several development and collaboration tools are also used during the implementation process. Git is utilized for version control to manage changes in the source code and track development progress. GitHub is used for project collaboration, code management, and maintaining repositories. Postman is employed for testing APIs and validating backend functionality before integration with the frontend system.

The project follows an iterative development methodology in which features are developed, tested, refined, and integrated continuously. This approach allows incremental improvements, easier debugging, and better management of project requirements throughout the development lifecycle.

1.6 Expected Outcome & Benefits

The expected outcome of TradeLearn is the development of a fully functional web-based platform that successfully enables users to exchange skills through a structured credit-based model. The platform aims to establish an interactive learning ecosystem where individuals can both teach and learn while removing traditional financial barriers associated with education. The system is designed to promote participation, encourage collaboration, and create a sustainable environment where knowledge sharing becomes a mutually beneficial activity.

The proposed platform is expected to provide several core functionalities that support smooth and efficient operation of the skill exchange process. These expected outputs include a user account management system that enables secure registration and login functionality, a skill exchange mechanism that allows users to connect and interact with one another, and a credit management system that maintains earned and utilized credits. Additionally, the system includes request handling features for managing exchange activities, dashboard implementation for tracking user progress and activities, and secure authentication mechanisms that ensure user privacy and data protection.

TradeLearn provides several benefits that contribute to both educational and technical perspectives. One of the major benefits of the platform is the promotion of collaborative learning. Instead of following traditional isolated learning methods where users only consume

educational content, the platform encourages interaction and participation among users. Through peer-to-peer engagement, learners gain opportunities to exchange experiences and learn through direct communication and collaboration.

Another significant benefit is accessible education. Many online learning platforms require expensive subscriptions and course fees, which limit access for many users. TradeLearn removes this financial dependency and allows individuals to acquire knowledge regardless of economic limitations by using a credit-based exchange system instead of monetary payments.

The platform also encourages community participation by transforming users from passive consumers into active contributors. Users not only learn from others but also participate by teaching skills and sharing expertise. This approach strengthens community engagement and creates a culture of mutual growth and support.

Transparency is another important benefit provided by the system. Features such as request tracking, learning history, activity records, and dashboard monitoring allow users to track their interactions and progress. This improves trust and accountability among platform users and creates a more organized exchange process.

TradeLearn also provides a practical learning environment where users acquire real-world knowledge directly from peers who possess relevant experience and expertise. Such interactive learning often provides greater understanding and practical insights compared to traditional content consumption approaches.

From a technical perspective, the project also serves as an implementation of various full-stack development concepts and technologies. The system demonstrates practical application of full-stack architecture, database integration, API development, authentication mechanisms, and system design principles. Therefore, apart from solving a real-world problem, the project also contributes significantly to technical learning and practical software development experience.

1.7 Original Contribution Statement

TradeLearn represents an original implementation developed specifically to address challenges associated with traditional learning platforms and limitations in skill accessibility.

Although the project uses standard technologies and open-source development frameworks, the overall concept, system workflow, database design, and implementation logic are independently created.

Key original contributions include:

Credit-Based Knowledge Economy

The platform introduces a mechanism where knowledge itself acts as currency.

Structured Peer-to-Peer Exchange

Instead of direct content consumption, users actively participate in skill transactions.

Customized Workflow Design

The request lifecycle and dashboard tracking system were uniquely designed.

Independent Database Architecture

Entity relationships and schema structure were designed specifically according to project requirements.

Integrated System Design

Authentication, skill exchange, requests, and credits are combined into one unified ecosystem.

The project demonstrates independent problem analysis, requirement identification, system design, implementation strategy, and feature integration.

Therefore, TradeLearn serves as an original educational technology solution supporting collaborative learning without relying on financial transactions.